Data Retrieval and Data Wrangling/Cleaning

Project sections

1. **Data Retrieval**
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   - Data acquisition and loading
   - Storage of raw data in original format
   - Basic validation of retrieved data
2. **Data Wrangling/Cleaning**
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   - Use of `quantDDA()` and `vizDDA()` for exploratory diagnostics
   - Data cleaning and preprocessing
   - Construction of a reproducible wrangling pipeline
   - Preparation of cleaned dataset for EDA and modeling

Data Source Documentation

1. Dataset Origin

The dataset used in this project is the **New York City Taxi & Limousine Commission (TLC) – High Volume For-Hire Vehicles (HVFHV) Trip Record Data**.

- Source: NYC Taxi & Limousine Commission (TLC)
- Dataset: High Volume For-Hire Vehicles (HVFHV) Trip Records
- Website: https://www.nyc.gov/site/tlc/about/tlc-trip-record-data.page
- Months used: December 2025 and January 2026
- Date accessed: February 24th, 2026

The TLC provides publicly available trip-level data for high-volume for-hire vehicle services (e.g., rideshare platforms operating under TLC regulation).

The dataset was accessed programmatically using the NYC Open Data API (Socrata) instead of direct file downloads. A custom data retrieval function was implemented to query the High Volume For-Hire Vehicle (HVFHV) dataset using HTTP requests.

The API was used to filter records by date range for December 2025 and January 2026, while selecting only the relevant features required for the analysis. Due to API constraints on response size, the data was retrieved in batches using pagination with limit and offset parameters. Each batch was appended and combined into a single DataFrame for each month.

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2. Dataset Description

The dataset contains trip-level records for HVFHV rides in New York City. Each row represents a single trip and includes:

- Request, pickup, and drop-off timestamps
- Trip distance
- Pickup and drop-off location IDs
- Dispatching and originating base numbers
- Shared ride indicators
- Other operational trip attributes

The combined December 2025 and January 2026 datasets contain tens of millions of trip records, making this a large-scale dataset appropriate for data-intensive analysis.

3. Data Format and Structure

The data was provided in Parquet format.

- Each row corresponds to one HVFHV trip.
- Each column represents a trip attribute.
- The dataset is structured as a large tabular dataset with both numerical and categorical variables.

Raw data files were stored in their original format before any transformations were applied to ensure reproducibility and traceability of preprocessing steps.

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4. Justification for Use

This dataset is appropriate for the project because:

- It satisfies the large-scale data requirement (millions of observations).
- It contains both numerical and categorical variables suitable for statistical analysis.
- It allows formulation of supervised learning problems (e.g., predicting trip duration or other continuous variables).
- It supports unsupervised analysis (e.g., clustering ride patterns by time or location).
- It represents real-world transportation data with practical analytical value.

The December 2025 and January 2026 data provide recent, consecutive monthly observations, enabling temporal comparison and pattern analysis.

1. Imports

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
import scipy.stats as stats
import pyarrow
import urllib.parse
import time
```

Reading the two raw data set and combining them to get a dataset > 1GB

```python
def load_month_api(start_date, end_date, limit_total=200000, chunk_size=50000):
    base_url = "https://data.cityofnewyork.us/resource/r67x-e97r.json"

    all_parts = []

    for offset in range(0, limit_total, chunk_size):
        query = {
            "$select": ",".join([
                "trip_miles",
                "trip_time",
                "base_passenger_fare",
                "tips",
                "driver_pay",
                "pickup_datetime"
            ]),
            "$where": f"pickup_datetime >= '{start_date}' AND pickup_datetime < '{end_date}'",
            "$limit": chunk_size,
            "$offset": offset
        }

        url = base_url + "?" + urllib.parse.urlencode(query)
        part = pd.read_json(url)

        if part.empty:
            break

        all_parts.append(part)
        time.sleep(0.5)  # avoid getting rate-limited

    df = pd.concat(all_parts, ignore_index=True)
    return df

    df_1 = load_month_api(
    "2025-12-01T00:00:00",
    "2026-01-01T00:00:00"
)

print(df_1.shape)


df_2 = load_month_api(
    "2026-01-01T00:00:00",
    "2026-02-01T00:00:00"
)

print(df_2.shape)
```

```python
# Combine
df = pd.concat([df_1, df_2], ignore_index=True)

print("Combined shape:", df.shape)

# Save combined version
df.to_parquet("processed_data/tripdata_2025_12_2026_01.parquet", index=False)

print("Combined file saved successfully!")
```

```
File 1 shape: (22108438, 25)
File 2 shape: (20940373, 25)
Combined shape: (43048811, 25)
Combined file saved successfully!
```

3. Information on Raw Data

Getting the information of the combined datasets

```
[ ]
df.info()
```

```
<class 'pandas.DataFrame'>
RangeIndex: 43048811 entries, 0 to 43048810
Data columns (total 25 columns):
 #   Column                Dtype
---  ------                -----
 0   hvfhs_license_num     str
 1   dispatching_base_num  str
 2   originating_base_num  str
 3   request_datetime      datetime64[us]
 4   on_scene_datetime     datetime64[us]
 5   pickup_datetime       datetime64[us]
 6   dropoff_datetime      datetime64[us]
 7   PULocationID          int32
 8   DOLocationID          int32
 9   trip_miles            float64
 10  trip_time             int64
 11  base_passenger_fare   float64
 12  tolls                 float64
 13  bcf                   float64
 14  sales_tax             float64
 15  congestion_surcharge  float64
 16  airport_fee           float64
 17  tips                  float64
 18  driver_pay            float64
 19  shared_request_flag   str
 20  shared_match_flag     str
 21  access_a_ride_flag    str
 22  wav_request_flag      str
 23  wav_match_flag        str
 24  cbd_congestion_fee    float64
dtypes: datetime64[us](4), float64(10), int32(2), int64(1), str(8)
memory usage: 8.6 GB
```

4. Processing Missing Values

Checking for missing values

```
df.isnull().sum().sort_values(ascending=False)
```

```
originating_base_num    11791229
hvfhs_license_num              0
dispatching_base_num           0
request_datetime               0
on_scene_datetime              0
pickup_datetime                0
dropoff_datetime               0
PULocationID                   0
DOLocationID                   0
trip_miles                     0
trip_time                      0
base_passenger_fare            0
tolls                          0
bcf                            0
sales_tax                      0
congestion_surcharge           0
airport_fee                    0
tips                           0
driver_pay                     0
shared_request_flag            0
shared_match_flag              0
access_a_ride_flag             0
wav_request_flag               0
wav_match_flag                 0
cbd_congestion_fee             0
dtype: int64
```

Missing Values Analysis: Calculating the percentage of missing values in originating_base_num

```
missing_pct = df["originating_base_num"].isnull().mean() * 100
print(f"{missing_pct:.2f}% missing")
```

```
27.39% missing
```

Since at this stage of the analysis we do not know if originating_base_num column is useful or not, we do not drop it. instead, we fill-in the missing values for this column with "UNKNOWN"

```
df["originating_base_num"] = df["originating_base_num"].fillna("UNKNOWN")
```

5. Processing Duplicates

Checking for duplicates and dropping the duplicate rows Since duplicate records can bias modeling results, we removed it.

```
df.duplicated().sum()
df = df.drop_duplicates()

print("New shape:", df.shape)
```

```
New shape: (43048810, 25)
```

5. validity constraints

- taxi trips cannot have negative values for trip miles, trip time, or base passenger fare

```
print("Trip miles <= 0:", (df["trip_miles"] <= 0).sum())
print("Trip time <= 0:", (df["trip_time"] <= 0).sum())
print("Base fare <= 0:", (df["base_passenger_fare"] <= 0).sum())
```

```
Trip miles <= 0: 5139
Trip time <= 0: 739
Base fare <= 0: 135730
```

```
# calculating the percentage of the non-valid negative values for different attributes
total = len(df)

print("Trip miles <= 0: {:.4f}%".format(5139 / total * 100))
print("Trip time <= 0: {:.4f}%".format(739 / total * 100))
print("Base fare <= 0: {:.4f}%".format(135730 / total * 100))
```

```
Trip miles <= 0: 0.0119%
Trip time <= 0: 0.0017%
Base fare <= 0: 0.3153%
```

```
# trip_miles ≤ 0: Trips cannot have 0 or negative miles --> removed
# trip_time ≤ 0: Trips cannot have 0 or negative duration --> Remove
df = df[(df["trip_miles"] > 0) & (df["trip_time"] > 0)]
print("New shape after removing invalid trips:", df.shape)
```

```
New shape after removing invalid trips: (43043670, 25)
```

```python
# base_passenger_fare ≤ 0: Could fare = 0 be: Cancelled trips or Special program or Data recording issue
print(df[df["base_passenger_fare"] <= 0].head())
df[df["base_passenger_fare"] <= 0][["trip_miles", "trip_time", "tips"]].describe()
```

```
       hvfhs_license_num dispatching_base_num originating_base_num  \
3557              HV0003                B03404                B03404
21878             HV0003                B03404                B03404
28334             HV0003                B03404                B03404
40566             HV0003                B03404                B03404
56014             HV0003                B03404                B03404

          request_datetime   on_scene_datetime     pickup_datetime  \
3557  2025-12-01 00:11:03 2025-12-01 00:16:58 2025-12-01 00:16:58
21878 2025-12-01 01:41:53 2025-12-01 01:43:44 2025-12-01 01:44:51
28334 2025-12-01 02:09:24 2025-12-01 02:11:56 2025-12-01 02:12:29
40566 2025-12-01 04:16:18 2025-12-01 04:18:38 2025-12-01 04:20:11
56014 2025-12-01 05:05:00 2025-12-01 04:57:21 2025-12-01 05:00:13

          dropoff_datetime  PULocationID  DOLocationID  trip_miles  ...  \
3557  2025-12-01 00:23:38           215           205        1.76  ...
21878 2025-12-01 01:47:36           169            94        0.59  ...
28334 2025-12-01 02:20:28            24            41        1.49  ...
40566 2025-12-01 04:27:14             3           185        1.74  ...
56014 2025-12-01 05:13:56           143           125        4.17  ...

       congestion_surcharge  airport_fee  tips  driver_pay  \
3557                   0.00          0.0   0.0        8.12
21878                  0.00          0.0   0.0        4.00
28334                  0.00          0.0   3.0        7.12
40566                  0.00          0.0   0.0        6.80
56014                  2.75          0.0   0.0       31.50

       shared_request_flag  shared_match_flag  access_a_ride_flag  \
3557                     N                  N                   N
21878                    N                  N                   N
28334                    N                  N                   N
40566                    N                  N                   N
56014                    N                  N                   N

       wav_request_flag  wav_match_flag cbd_congestion_fee
3557                  N               N                0.0
21878                 N               N                0.0
28334                 N               N                0.0
40566                 N               N                0.0
56014                 N               N                1.5

[5 rows x 25 columns]
```

	trip_miles	trip_time	tips
count	135466.000000	135466.000000	135466.000000
mean	4.419190	1093.180340	0.135424
std	6.254602	911.840274	1.125532
min	0.010000	9.000000	0.000000
25%	1.325000	544.000000	0.000000
50%	2.497000	877.000000	0.000000
75%	5.090000	1397.000000	0.000000
max	513.730000	47657.000000	54.590000

Some trips have zero base fare but positive distance and duration. These are likely promotional, adjusted, or contract-based rides. Therefore, they were retained in the dataset.

6. Feature Engineering

Creating a trip_duration feature using timestamps even though we have trip_time to verify consistency

```
df["calculated_trip_time"] = (
    df["dropoff_datetime"] - df["pickup_datetime"]
).dt.total_seconds()

# ensuring the trip times are values (positive values)
(df["calculated_trip_time"] <= 0).sum()
```

```
np.int64(1)
```

7. Dropping Invalid/Useless Rows/Columns and Downcasting

```
# Droping the single row where dropoff <= pickup (invalid duration)
df = df[df["calculated_trip_time"] > 0]

print("New shape:", df.shape)
```

```
New shape: (43043669, 26)
```

```
# Drop the helper column
df = df.drop(columns=["calculated_trip_time"])
```

We validated timestamp consistency by computing trip duration from pickup/dropoff. We removed the single record where dropoff time was not after pickup time.

Downcast Numeric Columns as righ now we are using more that 8 GB memory:

Convert float64 → float32

Convert int64 → int32 (if safe)

```
# Downcast floats
float_cols = df.select_dtypes(include=["float64"]).columns
df[float_cols] = df[float_cols].astype("float32")

# Downcast ints
int_cols = df.select_dtypes(include=["int64"]).columns
df[int_cols] = df[int_cols].astype("int32")

print("Memory after downcasting (GB):",
      df.memory_usage(deep=True).sum() / (1024**3))
```

```
Memory after downcasting (GB): 7.226738282479346
```

8. Saving Cleaned Data

Save the full cleaned Dataset

```
[ ]
df.to_parquet(
    "processed_data/tripdata_clean_full.parquet",
    index=False,
    compression="snappy"
)

print("Full cleaned dataset saved.")
```

Create Smaller EDA Subset (5M random samples)

```
[ ]
df_eda = df.sample(n=5_000_000, random_state=42)

print("EDA subset shape:", df_eda.shape)
```

Save EDA Subset

```
[ ]
df_eda.to_parquet(
    "processed_data/tripdata_clean_eda_5M.parquet",
    index=False,
    compression="snappy"
)

print("EDA subset saved.")
```

9. Quantitative Description

Quantitative Descriptions (quantDDA())

```
def quantDDA(df):
    stats_cols = [
        "num_observations",
        "num_entries",
        "num_unique",
        "num_missing",
        "num_outliers",
        "num_extreme_1pct",
        "modes",
        "mean",
        "std",
        "max",
        "min",
        "Q1",
        "Q2",
        "Q3",
        "skewness",
        "kurtosis"
    ]

    summary_df = pd.DataFrame(index=df.columns, columns=stats_cols)

    n_entries = len(df)

    for col in df.columns:
        s = df[col]

        # Stats that are valid for both numerical and nun-numberical values
        summary_df.loc[col, "num_entries"] = n_entries
        summary_df.loc[col, "num_observations"] = s.count()
        summary_df.loc[col, "num_missing"] = s.isna().sum()
        summary_df.loc[col, "num_unique"] = s.nunique(dropna=True)

        # Since there may be more than one mode, we store them as comma-separated string
        modes = s.mode(dropna=True)
        summary_df.loc[col, "modes"] = ", ".join(map(str, modes.tolist())) if len(modes) else ""

        # Put the default value for the other columns to be NaN, to be handle the non-numerical data which do not have the fallowing stats
        summary_df.loc[col, ["mean","std","max","min","Q1","Q2","Q3","skewness","kurtosis",
                             "num_outliers","num_extreme_1pct"]] = np.nan
```

```python
    # Calculating the stats that are only meaningful for the numberical data
    if pd.api.types.is_numeric_dtype(s):
        sn = s.dropna() # Dropping NaN values from the column
        if len(sn) > 0:
            summary_df.loc[col, "mean"] = sn.mean()
            summary_df.loc[col, "std"] = sn.std(ddof=1)
            summary_df.loc[col, "max"] = sn.max()
            summary_df.loc[col, "min"] = sn.min()

            q1 = sn.quantile(0.25)
            q2 = sn.quantile(0.50)
            q3 = sn.quantile(0.75)
            summary_df.loc[col, "Q1"] = q1
            summary_df.loc[col, "Q2"] = q2
            summary_df.loc[col, "Q3"] = q3

            # Outliers using IQR method
            iqr = q3 - q1
            lower = q1 - 1.5 * iqr
            upper = q3 + 1.5 * iqr
            summary_df.loc[col, "num_outliers"] = ((sn < lower) | (sn > upper)).sum()

            # Extreme values (bottom/top 1%)
            p01 = sn.quantile(0.01)
            p99 = sn.quantile(0.99)
            summary_df.loc[col, "num_extreme_1pct"] = ((sn < p01) | (sn > p99)).sum()

            # Skewness and Kurtosis (robust + consistent)
            summary_df.loc[col, "skewness"] = stats.skew(sn, bias=False)
            summary_df.loc[col, "kurtosis"] = stats.kurtosis(sn, fisher=True, bias=False)

return summary_df
```

```
quantDDA(df)
```

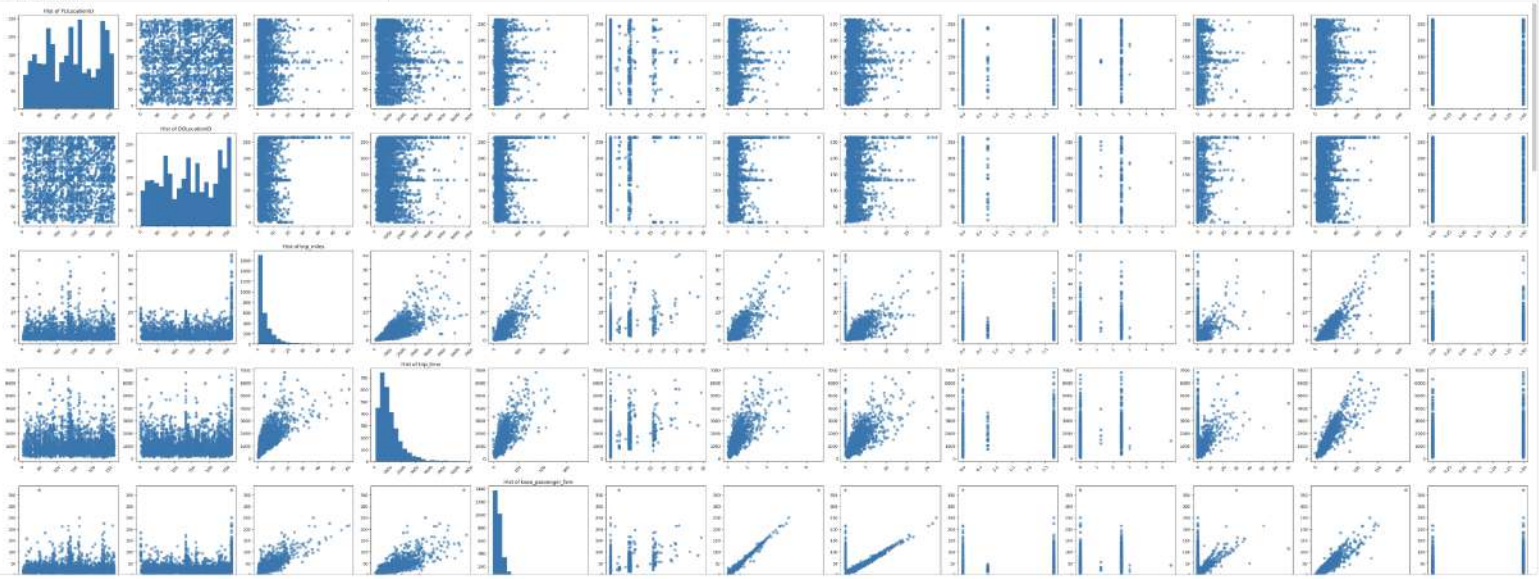
	num_observations	num_entries	num_unique	num_missing	num_outliers	num_extreme_1pct	modes	mean	std	max	min	Q1	Q2	Q3	skewness	kurtosis
hvfhs_license_num	43043669	43043669	2	0	NaN	NaN	HV0003	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN
dispatching_base_num	43043669	43043669	2	0	NaN	NaN	B03404	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN
originating_base_num	43043669	43043669	7	0	NaN	NaN	B03404	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN
request_datetime	43043669	43043669	5241639	0	NaN	NaN	2026-01-28 08:00:00	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN
on_scene_datetime	43043669	43043669	5251821	0	NaN	NaN	2025-12-12 20:09:42, 2025-12-13 19:39:25, 2025...	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN
pickup_datetime	43043669	43043669	5250654	0	NaN	NaN	2026-01-28 07:30:00	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN
dropoff_datetime	43043669	43043669	5251022	0	NaN	NaN	2026-01-24 23:57:47	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN
PULocationID	43043669	43043669	262	0	0	623489	138	137.77072	75.172581	265	1	74.0	138.0	208.0	-0.003661	-1.216052
DOLocationID	43043669	43043669	262	0	0	422445	265	141.308672	78.372052	265	1	75.0	140.0	216.0	-0.016777	-1.221058
trip_miles	43043669	43043669	62847	0	3346243	846815	1.1100000143051147	4.747509	5.670834	1454.430054	0.001	1.43	2.75	5.887	5.560913	342.877441
trip_time	43043669	43043669	11684	0	1909292	851569	502	1163.329871	842.759492	47657	1	576.0	940.0	1502.0	2.117862	11.116006
base_passenger_fare	43043669	43043669	40316	0	3181870	853767	8.9399995803833	26.726236	25.78771	1775.349976	-260.899994	11.65	18.82	31.76	4.034841	38.38327
tolls	43043669	43043669	2682	0	4473365	318532	0.0	1.032998	3.506727	108.410004	0.0	0.0	0.0	0.0	4.236968	21.267467
bcf	43043669	43043669	1911	0	3234869	771704	0.2199999988079071	0.663828	0.662491	47.209999	0.0	0.28	0.46	0.79	4.055892	38.783764
sales_tax	43043669	43043669	4263	0	3014222	429171	0.0	2.203025	2.093687	170.889999	0.0	0.95	1.58	2.7	3.332164	27.958968
congestion_surcharge	43043669	43043669	5	0	0	4	0.0	0.922642	1.294681	5.5	0.0	0.0	0.0	2.75	0.699576	-1.505388
airport_fee	43043669	43043669	17	0	3378939	76405	0.0	0.195896	0.675077	6.85	0.0	0.0	0.0	0.0	3.208966	8.623498
tips	43043669	43043669	7979	0	8527702	430276	0.0	1.255378	3.75784	223.25	0.0	0.0	0.0	0.0	5.791355	57.60598
driver_pay	43043669	43043669	28253	0	2397161	494191	4.0	20.633867	17.900681	1672.040039	-29.68	9.08	15.46	26.15	3.540672	40.853527
shared_request_flag	43043669	43043669	2	0	NaN	NaN	N	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN
shared_match_flag	43043669	43043669	2	0	NaN	NaN	N	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN
access_a_ride_flag	43043669	43043669	2	0	NaN	NaN	N	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN
wav_request_flag	43043669	43043669	2	0	NaN	NaN	N	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN
wav_match_flag	43043669	43043669	2	0	NaN	NaN	N	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN
cbd_congestion_fee	43043669	43043669	3	0	0	9	0.0	0.47719	0.698624	3.0	0.0	0.0	0.0	1.5	0.781	-1.390021

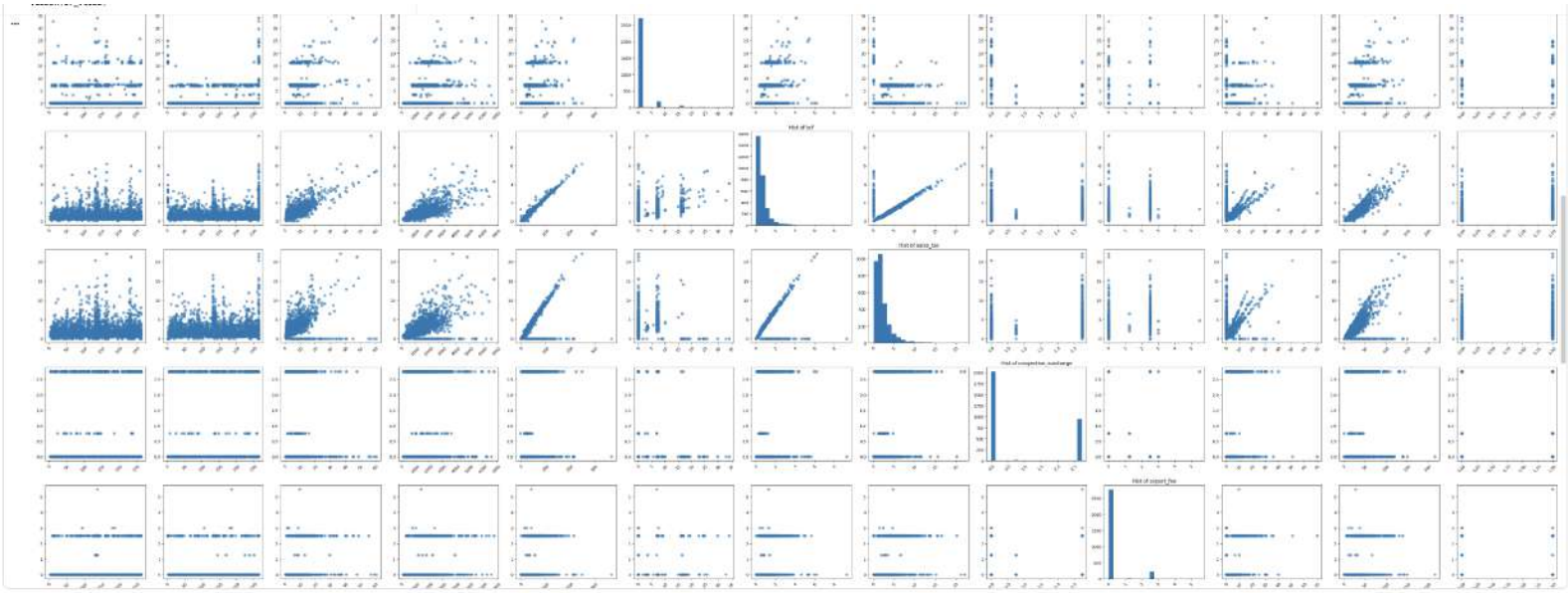
10. Visual Description

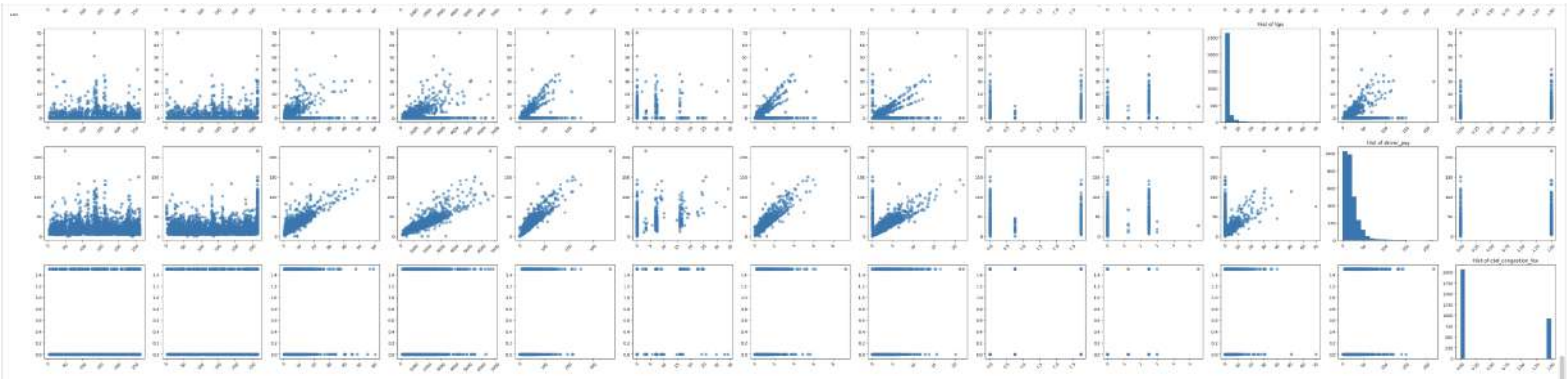
Visual Descriptions (vizDDA())

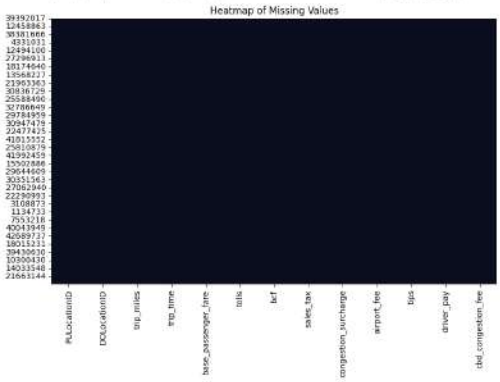
```python
def vizDDA(df):
    cols = df.columns
    n = len(cols)

    fig, axes = plt.subplots(n, n, figsize=(4*n, 4*n))

    for i in range(n):
        for j in range(n):
            x = cols[j]
            y = cols[i]
            ax = axes[i, j]

            # Diagonal: Univariate
            if i == j:
                if pd.api.types.is_numeric_dtype(df[x]):
                    ax.hist(df[x].dropna(), bins=20)
                    ax.set_title(f"Hist of {x}")
                else:
                    df[x].value_counts().plot(kind="bar", ax=ax)
                    ax.set_title(f"Bar of {x}")

            # Off-diagonal: Bivariate
            else:
                if pd.api.types.is_numeric_dtype(df[x]) and pd.api.types.is_numeric_dtype(df[y]):
                    ax.scatter(df[x], df[y], alpha=0.6)
                elif pd.api.types.is_numeric_dtype(df[x]) and not pd.api.types.is_numeric_dtype(df[y]):
                    sns.boxplot(x=df[y], y=df[x], ax=ax)
                elif not pd.api.types.is_numeric_dtype(df[x]) and pd.api.types.is_numeric_dtype(df[y]):
                    sns.boxplot(x=df[x], y=df[y], ax=ax)
                else:
                    sns.countplot(x=df[x], hue=df[y], ax=ax)

            ax.tick_params(axis='x', rotation=45)

    plt.tight_layout()
    plt.show()

    # Missing values heatmap
    plt.figure(figsize=(10, 6))
    sns.heatmap(df.isnull(), cbar=False)
    plt.title("Heatmap of Missing Values")
    plt.show()
```

```
# Since the dataset is very large (and after running the function on the whole dataset for more than 2 hour we did no  get any results, we ran this function on a subset of this dataset including 3000 random datas, and only the numerical features
df_vizDD = df.sample(n=3_000, random_state=42)
df_vizDD = df_vizDD.select_dtypes(include='number')

vizDDA(df_vizDD)
```




Hist of tvf
Hist of sales_tax
Hist of competitor_sustainge
Hist of export_fee


Hist of tips
Hist of driver_pay
Hist of del_competitor_tip


Heatmap of Missing Values
PULocationID
DOLocationID
trip_miles
trip_time
base_passenger_fare
tolls
bcf
sales_tax
congestion_surcharge
airport_fee
tips
driver_pay
cbd_congestion_fee